SIM **8.72** A CMOS cascode amplifier such as that in Fig. 8.34(a) has identical CS and CG transistors that have $W/L = 5.4\ \mu\text{m}/0.36\ \mu\text{m}$ and biased at $I = 0.2$ mA. The fabrication process has $\mu_n C_{ox} = 400\ \mu\text{A/V}^2$, and $V_A' = 5\ \text{V}/\mu\text{m}$. At what value of $R_L$ does the gain become $-100$ V/V? What is the voltage gain of the common-source stage?

**8.73** The purpose of this problem is to investigate the signal currents and voltages at various points throughout a cascode amplifier circuit. Knowledge of this signal distribution is very useful in designing the circuit so as to allow for the required signal swings. Figure P8.73 shows a CMOS cascode amplifier with all dc voltages replaced with signal grounds. As well, we have explicitly shown the resistance $r_o$ of each of the four transistors. For simplicity, we are assuming that the four transistors have the same $g_m$ and $r_o$. The amplifier is fed with a signal $v_i$.

(a) Determine $R_1$, $R_2$, and $R_3$. Assume $g_m r_o \gg 1$.
(b) Determine $i_1$, $i_2$, $i_3$, $i_4$, $i_5$, $i_6$, and $i_7$, all in terms of $v_i$. (*Hint:* Use the current-divider rule at the drain of $Q_1$.)
(c) Determine $v_1$, $v_2$, and $v_3$, all in terms of $v_i$.
(d) If $v_i$ is a 5-mV peak sine wave and $g_m r_o = 20$, sketch and clearly label the waveforms of $v_1$, $v_2$, and $v_3$.

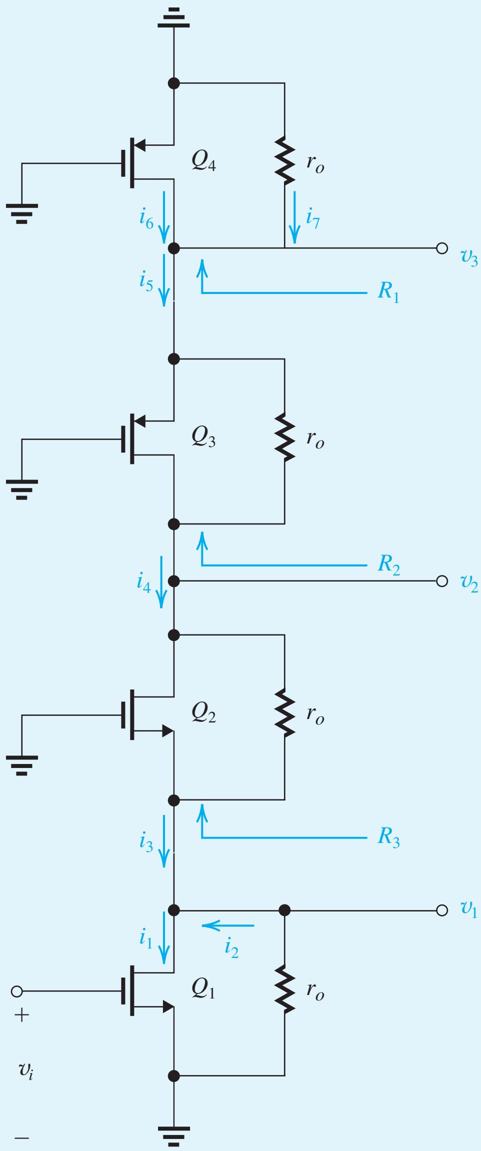


Figure P8.73

**D 8.74** Design the double-cascode current source shown in Fig. P8.74 to provide $I = 0.2$ mA and the largest possible signal swing at the output; that is, design for the minimum allowable voltage across each transistor. The 0.13-μm CMOS fabrication process available has $V_{tp} = -0.4$ V, $V_A' = -6\ \text{V}/\mu\text{m}$, and $\mu_p C_{ox} = 100\ \mu\text{A/V}^2$. Use devices with $L = 0.4\ \mu\text{m}$, and operate at $|V_{OV}| = 0.2$ V. Specify $V_{G1}$, $V_{G2}$, $V_{G3}$, and the $W/L$ ratios of the transistors. What is the value of $R_o$ achieved?

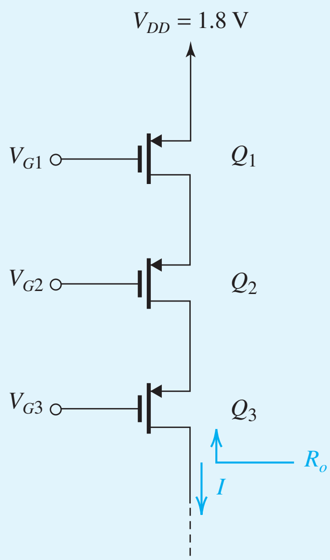


Figure P8.74

***8.75** Figure P8.75 shows a folded-cascode CMOS amplifier utilizing a simple current source $Q_2$, supplying a current $2I$, and a cascoded current source ($Q_4$, $Q_5$) supplying a current $I$. Assume, for simplicity, that all transistors have equal parameters $g_m$ and $r_o$.

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

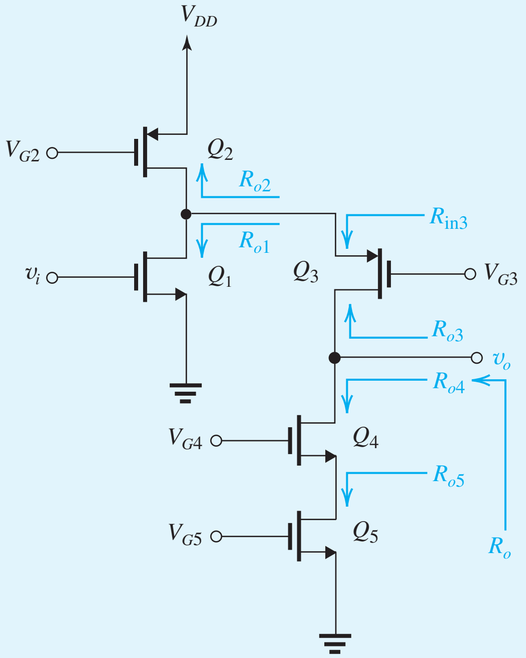


Figure P8.75

(a) Give approximate expressions for all the resistances indicated.
(b) Find the amplifier output resistance $R_o$.
(c) Show that the short-circuit transconductance $G_m$ is approximately equal to $g_{m1}$. Note that the short-circuit transconductance is determined by short-circuiting $v_o$ to ground and finding the current that flows through the short circuit, $G_m v_i$.
(d) Find the overall voltage gain $v_o/v_i$ and evaluate its value for the case $g_{m1} = 2$ mA/V and $A_0 = 30$.

**8.76** A cascode current source formed of two *pnp* transistors for which $\beta = 50$ and $V_A = 5$ V supplies a current of 0.2 mA. What is the output resistance?

**8.77** Use Eq. (8.88) to show that for a BJT cascode current source utilizing identical *pnp* transistors and supplying a current $I$,

$$IR_o = \frac{|V_A|}{(V_T/|V_A|) + (1/\beta)}$$

Evaluate the figure-of-merit $IR_o$ for the case $|V_A| = 5$ V and $\beta = 50$. Now find $R_o$ for the cases of $I = 0.1$, 0.5, and 1.0 mA.

**8.78** Consider the BJT cascode amplifier of Fig. 8.38 for the case all transistors have equal $\beta$ and $r_o$. Show that the voltage gain $A_v$ can be expressed in the form

$$A_v = -\frac{1}{2}\frac{|V_A|/V_T}{(V_T/|V_A|) + (1/\beta)}$$

Evaluate $A_v$ for the case $|V_A| = 5$ V and $\beta = 50$. Note that except for the fact that $\beta$ depends on $I$ as a second-order effect, the gain is independent of the bias current $I$!

**8.79** A bipolar cascode amplifier has a current-source load with an output resistance $\beta r_o$. Let $\beta = 50$, $|V_A| = 100$ V, and $I = 0.2$ mA. Find the voltage gain $A_v$.

**D *8.80** Figure P8.80 shows four possible realizations of the folded cascode amplifier. Assume that the BJTs have $\beta = 100$ and that both the BJTs and the MOSFETs have $|V_A| = 5$ V.

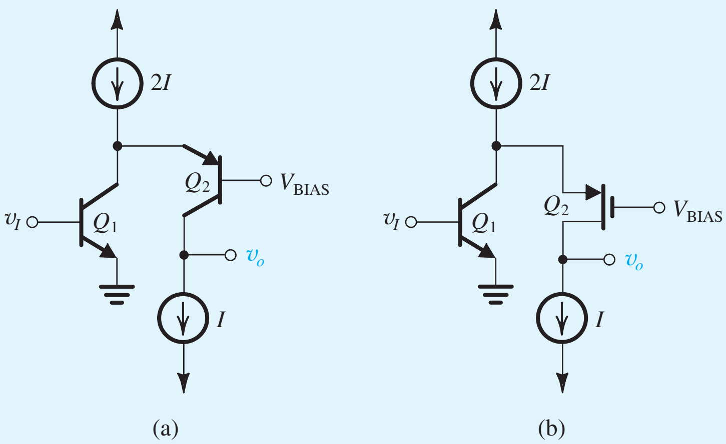


Figure P8.80